

Smart Tech For A Smart Grid

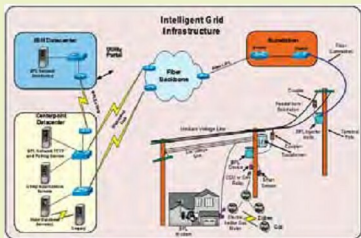
Integrated communications, sensing and measurement, intelligent devices, advanced control, improved interfaces and decision support...

A smart grid typically pervades the entire energy distribution system, right from power generation to power consumption. It is not a single technology but a combination of products, technologies, services and ecosystem partners that act at various levels to optimise communications, improve resilience, and reduce the operational cost and complexity of the energy grid.

"The smart grid can be thought of as comprising three layers: the physical power layer (transmission and distribution), the data transport and control layer (communications and control) and the applications layer (applications and services). All these layers have a set function to ensure that the grid is able to meet the challenges we see today and make our future better," says Vivek Tyagi, country sales manager, Freescale Semiconductor. And each level has various technologies at work.

Consider a sample of the technologies that fuel the smart grid:

- Communication network is the heart of the smart grid. State-of-the-art communication tools such as rugged integrated services routers (ISRs), hardened catalyst switches, and integrated security and software services are used across the network, while communication protocols ranging from radio frequency and Zigbee to optic fibre and GSM/GPRS act as the backbone of the network.
- Data centres play an important role in sharing and segmenting appropriate information across the fabric to help optimise energy distribution.
- Technologies that augment power from various sources before distribution help harness renewable energy sources such as solar and wind power.
- Intelligent monitoring systems keep track of the electricity flowing in the system.
- Power system automation tools use artificial intelligence and analytics to quickly diagnose and solve grid failures and power outages.
- Superconductive transmission lines reduce power loss.
- Automation devices such as fault-passage indicators, auto-reclosures, sectionalisers (load-break switches) and automated ring main units improve the efficiency of the system.
- Smart meters, combined with an automated metering infrastructure, help understand patterns in energy consumption (during peak-hours and non-peaks hours, in various localities and so on). Powered by two-way digital communication links, these also help control appliances in the consumer's household or machines on a factory floor; non-critical equipment can be shut down during peak times and restarted during lean times.
- Embedded control and integrated connectivity is at the heart of a true smart grid. A range of products such as low-



Smart grid architecture (courtesy: IBM)

INTELLIGENT GRID BUILDING BLOCKS	HOME INFORMATION AND CONTROL	HOME ENERGY MONITOR, HOME ENERGY CONTROL
	HOME AREA NETWORK	ZIGBEE SMART ENERGY PROFILE
	METERING DEVICES	ELECTRICITY METERS - TOU, 4-QUADRANT, REMOTE DISCONNECT, REVENUE PROTECTION, OUTAGE DETECTION, PROFILING
	WIDE-AREA NETWORK	GSM/GPRS, MESH RADIO, LONG-WAVE RADIO, DLG
KNOWLEDGE AREA	NETWORK SWITCHES	CAPACITOR BANK SWITCHES, RECONFIGURATION SWITCHES
	BUSINESS SYSTEMS	ENERGY ACCOUNTING, NETWORK MAPPING, BILLING, CRM, OMS, DSM etc.

Smart grid building blocks (courtesy: Secure Meters Limited)

power 8-bit microcontrollers (MCUs) for gas, water and heat meters; highly-integrated 32-bit MCUs for single- and three-phase electric meters; digital signal controllers (DSCs) for power line modems; 32-bit processors for energy gateways, broadband routers and concentrators; acceleration sensors for tamper detection; air-flow or liquid-flow pressure sensors for gas, water or heat meters; and ZigBee transceivers combined with MCUs for home area networks are some of the unseen technologies that drive the smart grid!

Almost all these technologies and products are available in India, though not yet put to full use. S. Ramesh, retired chief engineer-electricity, Karnataka Power Transmission Corporation Ltd, and consultant to the Central Power Research Institute (CPRI), who is now working on a pilot smart grid project, informs, "Smart meters are available from SEMS, Larsen & Toubro, etc. Other smart grid devices such as communication devices, intelligent electronic devices (IEDs), automation devices, and control and protection devices are available from ABB, Siemens, Schneider, Cisco, Acreva, Easun Reyrolle and many other companies."